

Unit – 1 Logic Gates and Combinational circuits

Logic Gates:- Logic gates is an electronic device that implement an elementary logic function such that AND, NOT, AND, NAND, NOR, Ex- OR. Each gate has a logic symbol that gives its logic function but does not indicate the electronics ckt. The use of these symbols simplifies the understanding of a complex logic ckt.

This logic gates represents the two logic states, logic 1 or logic 0. The logic state is the logic sense, i.e. true or false. These logic states are held at different voltage levels. That will be depends on the logic family. A logic family is is a range of electronic device that is made by same manufacturing technique & provides a no. of logic functions such that TTL, ECL, I²L, etc.

OR Gate:- OR gate is an electronic logic gate whose O/P is 0 when all the I/Ps of OR gate are 0, other wise 1. The symbol, Logical expression & truth table for this is as shown below.

Logic symbol



Logical expression:- $y = A + B$

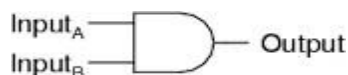
Truth table

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

This gate can be constructed in IC 7432. It is 14 pin IC. In this 4 OR gates are designed in one IC.

AND Gate:- AND gate is an electronic logic gate whose O/P is 1 when all the I/Ps of AND gate are 1, other wise 0. The symbol, Logical expression & truth table for this is as shown below.

Logic symbol



Logical expression:- $y = A * B$

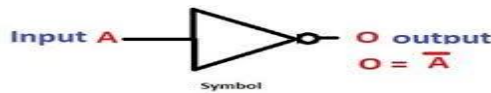
Truth table

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

This gate can be constructed in IC 7408. It is 14 pin IC. In this 4 AND gates are designed in one IC.

NOT Gate:- Not gate is electronic logic gate whose O/P is negation or compliment of I/P. i. e. When I/P is 0 then O/P is 1 & when I/p 0 then O/P is 0. It is also known as inverter or complimentary gate. The symbol, Logical expression & truth table for this is as shown below.

Logic symbol



Logical expression:- $y = A$

Truth table

A	Y
0	1
1	0

This gate can be constructed in IC 7404. It is 14 pin IC. In this 6 NOT gates are designed in one IC.

NOR Gate:- NOR gate is an electronic universal logic gate whose O/P is 1 when all the I/Ps of NOR gate are 0 , other wise 0. From this gate we can construct all other gates. The symbol, Logical expression & truth table for this is as shown below.

Logic symbol



Logical expression:- $y = A + B$

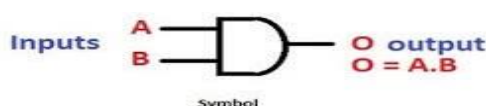
Truth table

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

This gate can be constructed in IC 7402. It is 14 pin IC. In this 4 OR gates are designed in one IC.

NAND Gate:- NAND gate is an electronic universal logic gate whose O/P is 0 when all the I/Ps of AND gate are 1, other wise 0. The symbol, Logical expression & truth table for this is as shown below.

Logic symbol



Logical expression:- $y = A * B$ Truth table

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

This gate can be constructed in IC 7400. It is 14 pin IC. In this 4 NAND gates are designed in one IC.

EX – OR (X – OR Gate):- Ex – OR gate is a exclusive electronic logic OR gate. The property of this gate is that the O/P is high only if odd no. of I/Ps are high, otherwise 0. The symbol, Logical expression & truth table for this is as shown below.

Logic symbol



Logical expression:- $y =$

$A B + A B$ Truth table

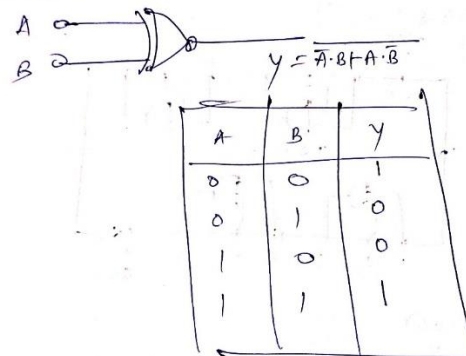
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

This gate can be constructed in IC 7486. It is 14 pin IC. In this 4 X – OR gates are designed in one IC.

EX – NOR (X – NOR Gate):-

Ex-NOR gate IC 74LS266, 74x266

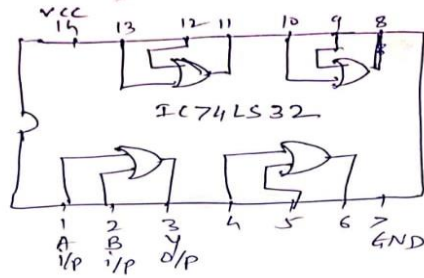
Ex NOR :-



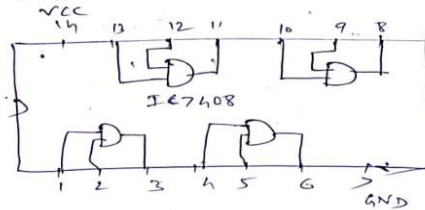
Pin connection of logic gates

Pin Configuration of logic gates

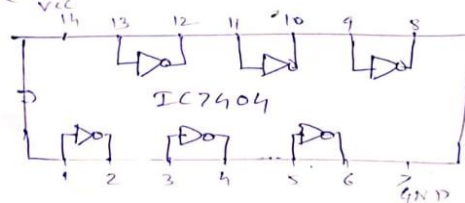
* OR gate (7432)



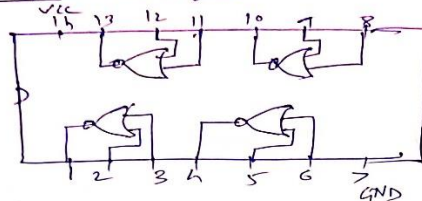
* AND gate (7408)



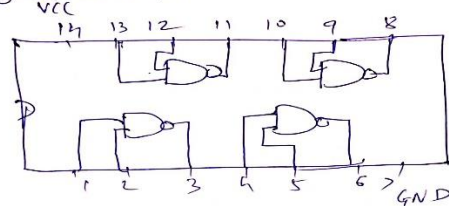
* NOT gate (7404)



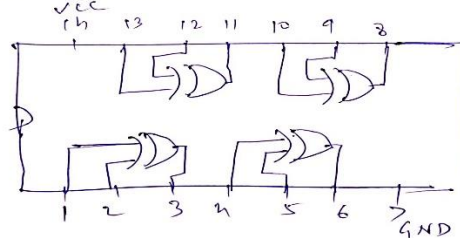
* IC 7402 NOR gate



* NAND gate IC 7400

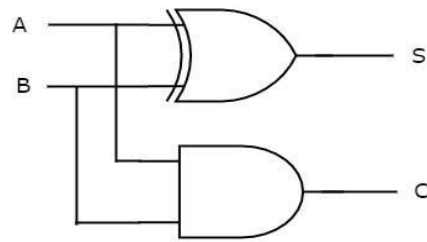


* XOR gate IC 7486



Application of Logic gates :

HALF ADDER:- A half adder is used to implement addition of 2 binary digits. It is not a complete adder as it is not able to add carries generated from the previous addition. The logic diagram for half adder is as shown in fig. below. It consists of one X-OR gate & one AND gate. The X – OR gate are used for addition & AND gate are used for carry.



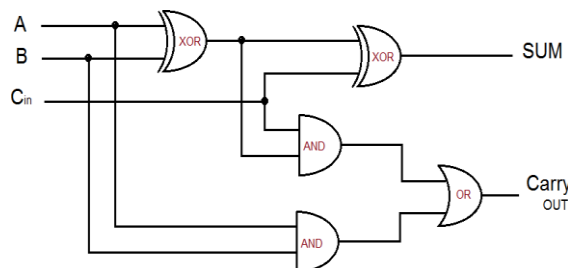
The working of half adder logic ckt. is as shown in following truth table.

A	B	Sum	Carry
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Full Adder :-

Any addition is said to be complete only when 2 bits along with carry generated are added. Half adder is incapable of doing this because it can add only two bit. Thus a full adder circuit required. The logic diagram for Full adder is as shown in fig. below.

The logic diagram consists of two X – OR gates which are used for addition of two bit & previous

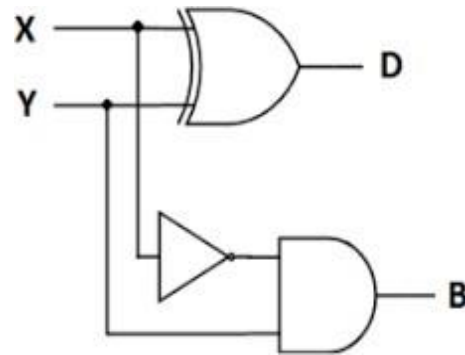


condition carry. Two AND gates are used & one OR gate are used for carry. The working of full adder is as shown in following truth table.

A	B	Cn	Sum	Carry
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

HALF SUBTRACTOR:- A half subtracter subtracts two bits. It is not a complete subtracter as it is unable to consider borrow generated from previous subtractions. Consider the binary subtraction of 2 bits. The logic ckt. Consists of EXOR gate for Difference, AND gate for barrow. But I/P for and gate is compliment of A. the logic diagram and its truth table for half subtractor is as below

Logic Diagram

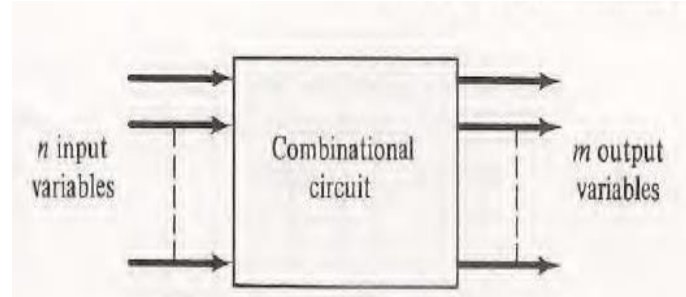


Truth table

A	B	Difference	Barrow
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Combinational Circuits:-

A combinational circuit is a connected arrangement of logic gates with a set of inputs and outputs. At any given time, the binary values of the outputs are a block diagram function of the binary combination of the inputs. A block diagram of a combinational circuit is shown below.

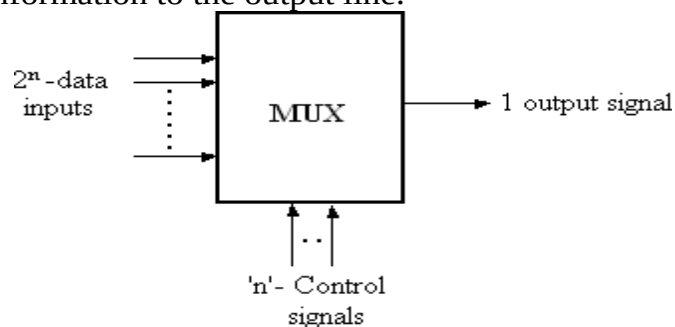


The n binary input variables come from an external source, the m binary output variables go to an external destination, and in between there is an interconnection of logic gates. A combinational circuit transforms binary information from the given input data to the required output data.

Multiplexers:- A multiplexer is known as many inputs and only one output.

A **multiplexer** or **MUX**, is a combinational circuit with more than one input line, one output line and more than one selection line.

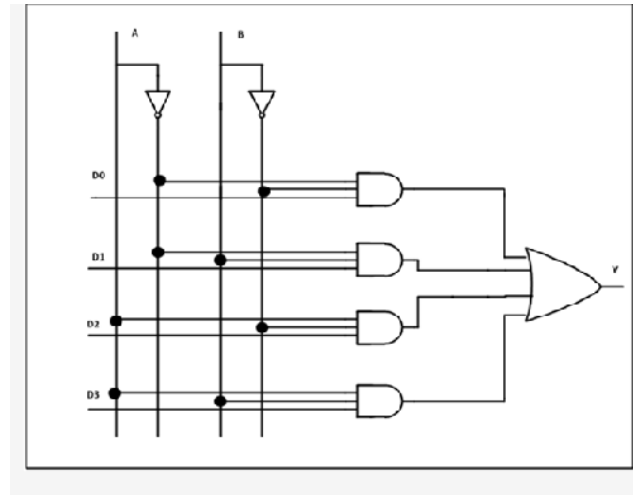
A multiplexer is also called a **data selector**, since it selects one of many inputs and steers the binary information to the output line.



There are different types of multiplexer

1. 4 to 1 multiplexer
2. 8 to 1 multiplexer

4 to 1 multiplexer :- The 4-to-1 multiplexer has 4 input bit, 2 control bits, and 1 output bit. The four input bits are D0,D1,D2 and D3. only one of this is transmitted to the output y. The output depends on the value of AB which is the control input. The control input determines which of the input data bit is transmitted to the output. For instance, as shown in fig. when AB = 00, the upper AND gate is enabled while all other AND gates are disabled. Therefore, data bit D0 is transmitted to the output, giving Y = D0.



The working of 4 to 1 multiplexer is as shown if following truth table

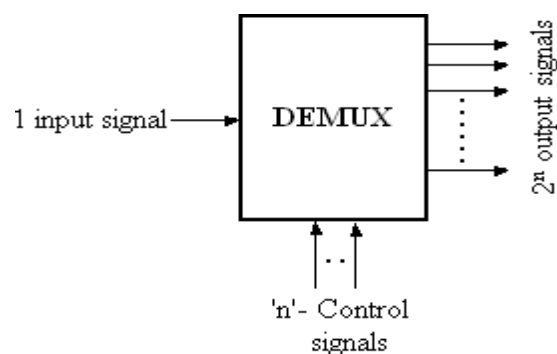
A	B	Y
0	0	D0
0	1	D1
1	0	D2
1	1	D3

Applications of Multiplexer: Multiplexer are used in various fields where multiple data need to be transmitted using a single line.

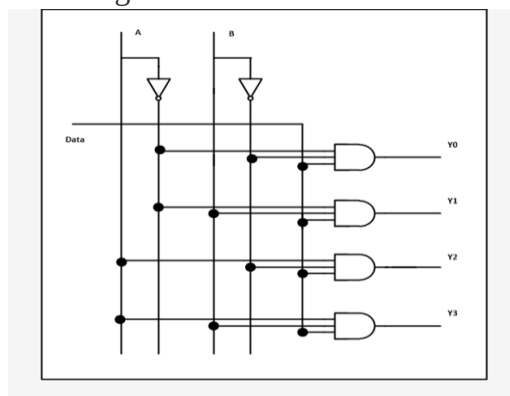
1. Communication system
2. Telephone network
3. Computer memory
4. Transmission from the computer system of a satellite

Demultiplexer:- Demultiplexer means one to many. A demultiplexer is a circuit with one input and many outputs. By applying control signal, we can steer any input to the output. Few types of demultiplexer are 1-to-2, 1-to-4, 1-to-8 and 1-to-16 demultiplexer.

Following figure shows the general idea of a demultiplexer with 1 input signal, m control signals, and n output signals.



1 to 4 Demultiplexer: The 1-to-4 demultiplexer has 1 input bit, 2 control bit, and 4 output bits. The 1-to-4 demultiplexer is shown in figure below-



The input bit is labeled as Data D. This data bit is transmitted to the data bit of the output lines. This depends on the value of AB, the control input.

When AB = 01, the upper second AND gate is enabled while other AND gates are disabled. Therefore, only data bit D is transmitted to the output, giving Y1 = Data.

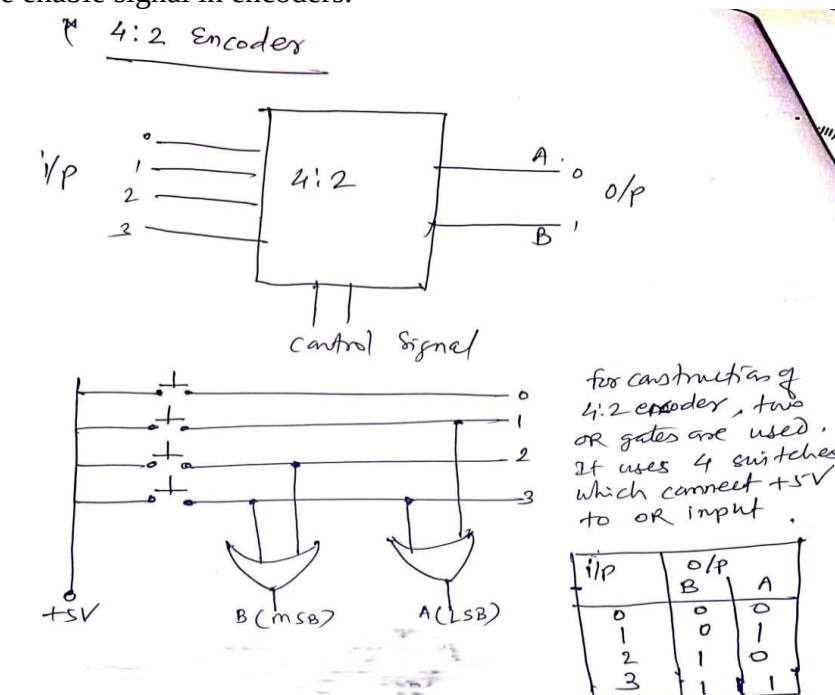
If D is low, Y1 is low. If D is high, Y1 is high. The value of Y1 depends upon the value of D. All other outputs are in low state.

If the control input is changed to AB = 10, all the gates are disabled except the third AND gate from the top. Then, D is transmitted only to the Y2 output, and Y2 = Data.

S1	S0	O/P(I/P comes at)
0	0	Y0
0	1	Y1
1	0	Y2

Applications of Demultiplexer: 1. Communication System 2. ALU (Arithmetic Logic Unit)
3. Serial to parallel converter

Encoder:- An **Encoder** is a combinational circuit that performs the reverse operation of Decoder. It has maximum of 2^n input lines and 'n' output lines. It will produce a binary code equivalent to the input, which is active High. Therefore, the encoder encodes 2^n input lines with 'n' bits. It is optional to represent the enable signal in encoders.

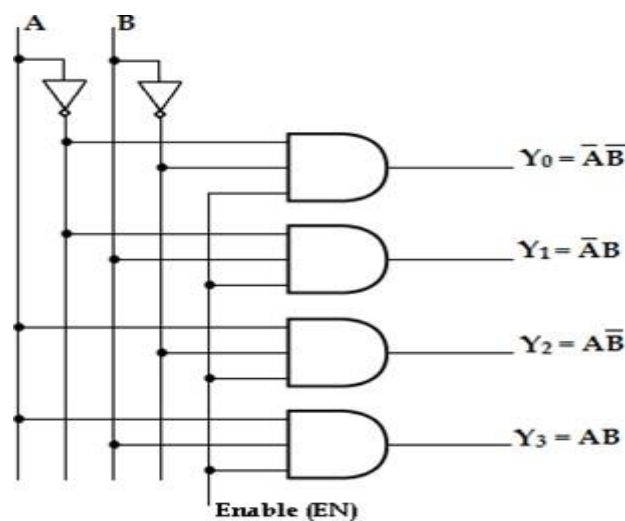


Decoders:- A decoder is a combinational circuit that converts binary information from n input lines to a maximum of 2^n unique output lines. The general structure of decoder circuit is



The encoded information is presented as n inputs producing 2^n possible outputs. The 2^n output values are from 0 through $2^n - 1$. A decoder is provided with enable inputs to activate decoded output based on data inputs. When any one enable input is unasserted, all outputs of decoder are disabled.

Binary Decoder (2 to 4 decoder): A binary decoder has n bit binary input and a one activated output out of 2^n outputs. A binary



Flip Flop

Sequential Circuit is a electronic logic ckt. Flip is Bistable Electronics Circuits.

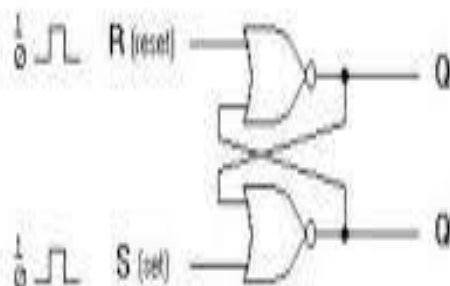
Bistable Means two States Set(1) Reset (0).Whose O/P is depends on the current I/P states as well as the previous O/P. Due to this concept we have to store the previous state O/P. They make extensive use of Flip – Flops. These ckts. are used in computers & data acquisition system in form of counter & registers. There are following different types of flip – flops.

1. RS F/F
2. JK F/F
3. MS JK F/F
4. D F/F
5. T F/F

RS F/F :-

RS F/F can be constructed by using two ways like Using NOR gate & by using NAND gate.

A. RS F/F/ by using NOR Gate: - The logic diagram for RSF/F by using NOR gate is as shown below.



The working for RS F/F by using NOR gate is divides into different four steps which is as shown in following truth table.

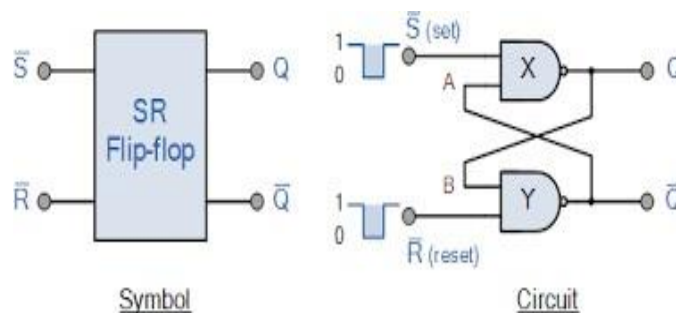
R	S	Q	Q	Condition
0	0	Previous O/P	Previous O/P	No change
0	1	1	0	SET
1	0	0	1	RESET
1	1	Not Allowed	Not Allowed	Race

In first step there is a no change in pervious O/P condition

In second step O/P of F/F goes in stable state, means in SET condition. In third step O/P goes in unstable state, means in RESET condition.

In fourth step O/P is not allowed condition. Because Both the O/P are logic 0.

B. RS F/F/ by using NAND Gate: - The logic diagram for RSF/F by using NAND gate is as shown below.



The working for RS F/F by using NAND gate is divides into different four steps which is as shown in following truth table.

S	R	Q	Q	Condition
0	0	Not Allowed	Not Allowed	Race
0	1	1	0	SET
1	0	0	1	RESET
1	1	Previous O/P	Previous O/P	No change

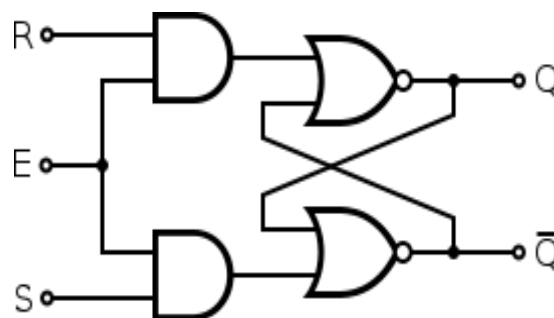
In first step O/P is not allowed condition. Because Both the O/P are logic 0. In

second step O/P of F/F goes in stable state, means in SET condition.

In third step O/P goes in unstable state, means in RESET condition. In

fourth step there is a no change in pervious O/P condition **Clocked RS F/F**

The logic diagram for clocked RS F/F by using NAND gate is as shown in fig. below.

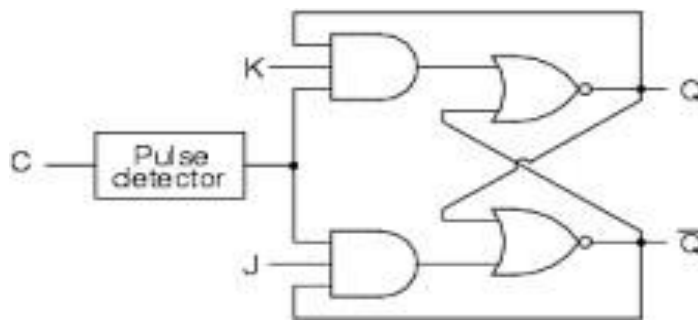


The RS F/F ckt. works as a regular only when clock is high. When clock goes low then ckt. does not work, means it will give previous state O/P. The truth table for this is as shown below table.

Clock	R	S	Q	Q	Condition
1	0	0	PREVIOUS O/P	PREVIOUS OUTPUT	Race
1	0	1	1	0	SET
1	1	0	0	1	RESET
1	1	1	NOT ALLOWED	NOT ALLOWED	No change
0	x	x	Previous O/P	Previous O/P	No change

J – K Flip-flop:- RS flip-flops gives one important drawback if suppose R and S inputs are gives same (i.e. 0 or 1). Then that ckt. Gives race condition. To eliminate this condition and to make the flip-flops function even if inputs are same, the RS flip-flop is modified to get the JK flip-flop.

The logic diagram for J K F/F is as shown fig. below. In the J-K flip-flop the output Q is fed as an additional input to gate A and Q is fed as an additional input to gate B apart from J-K inputs and clock.



The working of JK F/F divides in to four steps, which is as shown in following truth table with its logic symbol.

Clock	J	K	Q
1	0	0	Previous State
1	0	1	0(Reset)
1	1	0	1 (Set)
1	1	1	Q Toggling
0	x	x	Previous State

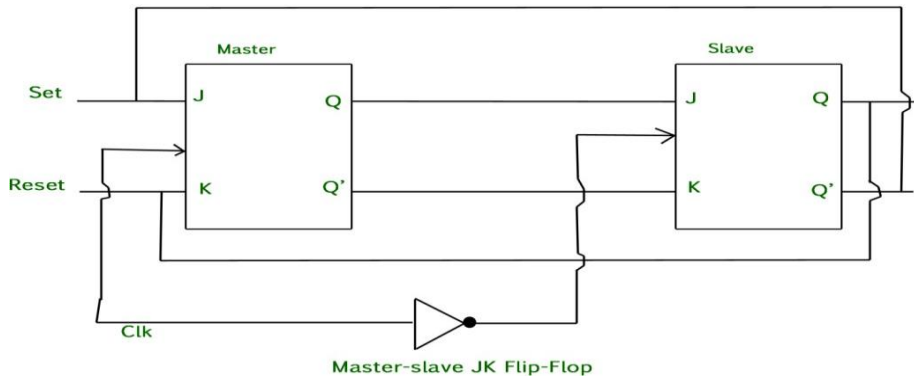
J K MASTER SLAVE FLIPFLOPS

The two J - K flip-flops are cascaded in MS JK flip flops. One works as master, the other is a slave which follows the master always. While the master is accepting the new inputs, slave is busy maintaining previous outputs as it is. When the master has completed accepting the inputs the slave will allow the changes caused by the master to reach the output. During this time no new inputs are accepted by the master.

When clock pulse = 1 master accepts new input and causes changes.

When clock pulse = 0 slave allows the changes to come at the output but master doesn't accept any new input

Example of J K Master slave flip-flop is 1C 7473 (dual JK flip-flop), 1C 7476. The logic symbol diagram for MS JK F/F is as shown below with logic gates.



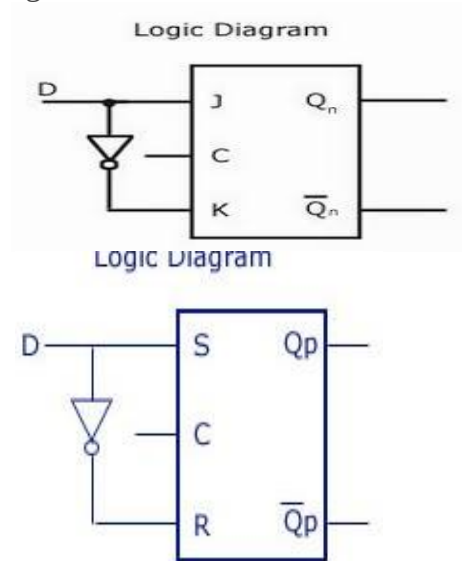
D (Delay) FLIP-FLOPS

If we consider only the following two conditions of a SR or a JK flip-flop.

$S = 1$	$R = 0$	$Q = 1$
$S = 0$	$R = 1$	$Q = 0$

$J = 1$	$K = 0$	$Q = 1$
$J = 0$	$K = 1$	$Q = 0$

The D F/F by using RS/ JK F/F is as shown below.



In D F/F S and R inputs or J and K inputs can be inverted. This common input is called the D input and this configuration is the D flip flop.

When $D = 1$ i.e. $S = 1, R = 0$ or $J = 1, K = 0$. then $Q = 1$.

When $D = 0$ i.e. $S = 0, R = 1$ or $J = 0, K = 1$, then $Q = 0$.

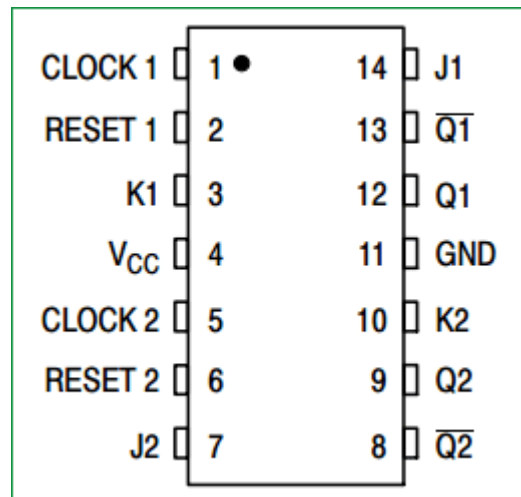
Q will always follow D i/p with the clock, after some time delay. D flip-flop is also called the delay flip-flop. A D flip-flop is used as a 1 bit latch or memory. For storing several bits, many D flip-flops are cascaded to form a register.

IC 7475 (D flip-flops)

It is a 16 pin DIP TTL IC working on 5V supply. It is a Quad (4) bistable latch. It uses MSI technology (Medium Scale Integration). It is ideal for handling 4 bits of data.

It has 4 inputs (D type), 8 outputs and 2 clocks. One clock synchronizes two flip-flops. It has 2 pins reserved for supply and ground. This IC is a positive level triggered latch. It is also called transparent latch. The reason is as long as clock = 1, all variations of D are passed on to the output.

When clock = 0 the last D input is latched at the output. The pin function diagram for this is as shown below

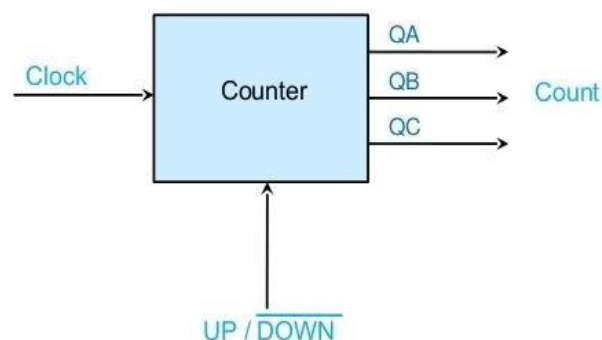


Counter :

These are logic circuits used to count clock pulses. They are basically electronic counting systems. Also counter can be used for frequency division and sequencing.

In a counter, with each clock pulse the output of the counter is incremented by 1. The counting could be in BCD or binary. For counter J K flip-flops used only in toggle mode. The number of flip-flops decides the maximum events (clock pulses) that can be counted by a counter.

A single flip flop can count 2 states 1 and 0, 2 flip flops can count 00, 01, 10, 11 i.e. 4 states. So in general n flip flops can count 2^n states/clock pulses. A counter can be shown with a general block diagram as



There are following types of counter.

- 1 Asynchronous counters
2. Synchronous counters

Further each counter can be sub classified as

1. Up counter
2. Down Counter

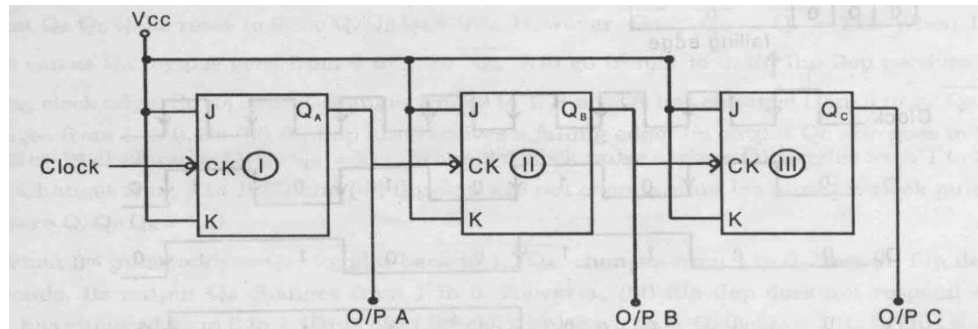
The up counter counts upwards with each clock pulse i.e. 00 → 99.

The down counter counts downwards with each clock pulse. The initial count is the biggest one, i.e., 99 - 00.

1. ASYNCHRONOUS COUNTER:- This is also called a ripple counter. The clock is connected only to the first flip-flop. The output of the 1st flip flop acts as clock to the second flip-flop. The output of

2nd flip-flop acts as clock to the third and so on. Thus second flip-flop can respond only after the first. Third flip flop can respond only after 1st and 2nd and so on. So we find that it is very slow. These counters are used for low to medium frequency applications. However they are comparatively easy in construction using less hardware and have simple operation.

1. **3 bit Asynchronous up counter:-** The logic diagram for 3 bit Asynchronous up counter is as shown below.

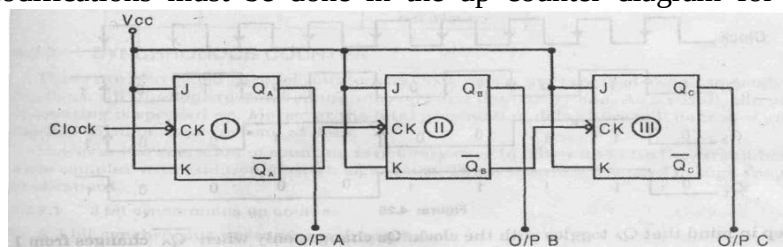


The functioning can be summarized with a truth table and a timing diagram. Also note that counting begins from 000 and moves till 111, which is nothing, but up counting.

C	B	A	Counted state
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7
0	0	0	0

3 bit Asynchronous down counter

The following modifications must be done in the up counter diagram for getting a downcounter.



The working can be summarized with a truth table and timing diagram

C	B	A	Counted state
0	0	0	0
1	1	1	7
1	1	0	6
1	0	1	5
1	0	0	4
0	1	1	3
0	1	0	2
0	0	1	1
0	0	0	0

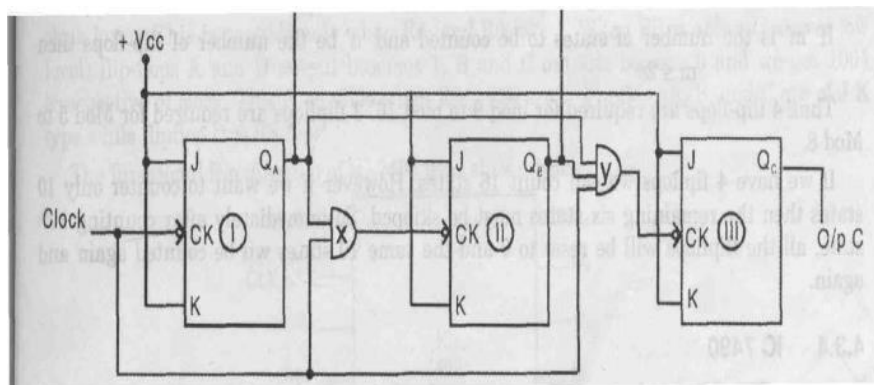
2. **SYNCHRONOUS COUNTER:-** These are also called parallel counters as clock pulses are

provided simultaneously to all flip-flops. All flip-flops respond simultaneously and not one by one. As a result, the process of counting is speeded up. Moreover the total propagation delay here will be that of any one flip-flop as all of them respond at once.

However the operation of counting is not very easy to follow and also the circuit becomes more complex with addition of extra logic gates. These counters are used in high frequency applications.

3 bit synchronous up counter:- A 3 bit synchronous up counter is drawn below

Working of the above circuit can be explained as follows.



Flipflop (I) toggles with every clock pulse. Flipflop (II) toggles with both clock AND output QA . flipflop (III) toggles with clock AND output QA AND Output QB.

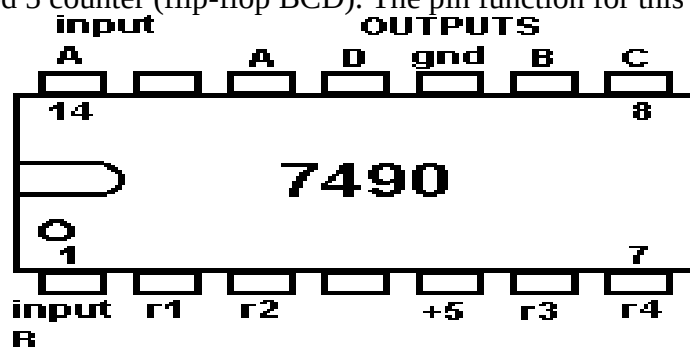
The functioning can be summarized with a truth table and a timing diagram. Also note that counting begins from 000 and moves till 111, which is nothing, but up counting.

C	B	A	Counted state
0	0	0	0
0	0	1	1
0	1	0	2
0	1	1	3
1	0	0	4
1	0	1	5
1	1	0	6
1	1	1	7
0	0	0	0

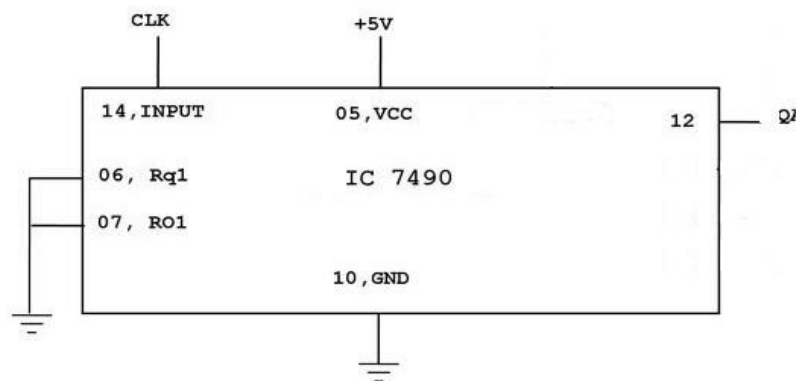
IC 7490:- It is a TTL(**Transistor Transistor Logic**) IC working on 5V power supply.

It is a Mod 10 counter. It can count a maximum of 10 states. It is a BCD Counter.

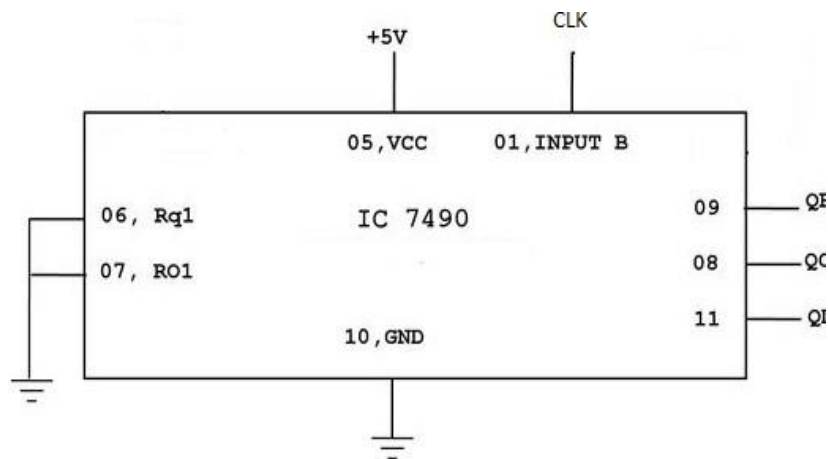
It is 14 pin IC in DIP works on +5v power supply. Internally there are 4 flip-flops ABC and D. Flip-flop A count 2 states 0 and 1, so it will act as mod 2 counter. Flip-flop B, C and D are connected together but only to count 5 states. So it will be a mod 5 counter. When all the four flip-flops are connected $2 \times 5 = 10$ states can be counted. There is a clock input to the mod 2 counter (flip-flop A) and mod 5 counter (flip-flop BCD). The pin function for this is as shown below.



MOD -2(Divide-by-2) Counter:- To produce a standard divide-by-2 counter, we can disable the other F/F(BCD) and apply the clock input signal directly to pin 1 (CLK_A) with the output signal being taken from pin 12 (Q_A) as shown.

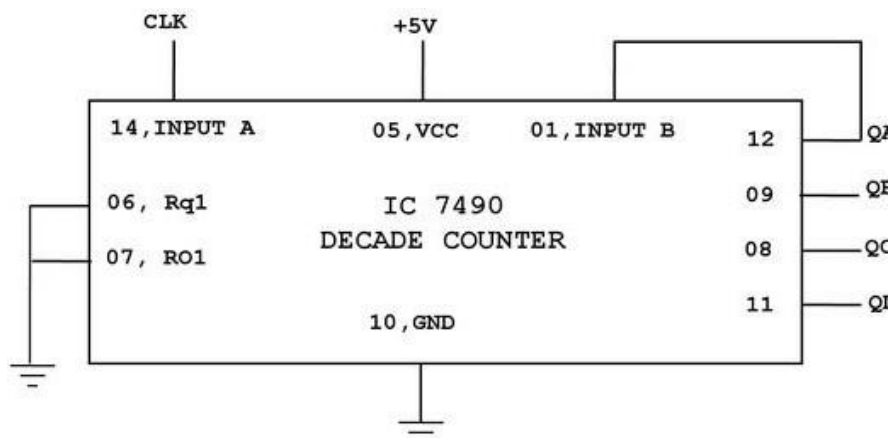


MOD- 5(Divide-by-5 Counter) To produce a standard divide-by-5 counter, we can disable the first flip-flop above, and apply the clock input signal directly to pin 1 (CLK_B) with the output signal being taken from pin 8,9 11 (Q_D, Q_C, Q_B) as shown.



MOD- 10(Divide-by-10 Counter):- To produce a divide-by-10 BCD decade counter, both internal counter circuits are used giving a 2 times 5 divide-by value. Since the first output Q_A from flip-flop “A” connected to F/F BCD CLK_B input as shown.

Then we can see that BCD counters are binary counters that count from 0000 to 1001 and then resets as it has the ability to clear all of its flip-flops after the ninth count. If we connect a pushbutton switch (SW₁) to clock input CLK_A, each time the pushbutton switch is released the counter will count by one. If we connected light emitting diodes (LED’s) to the output terminals, Q_A, Q_B, Q_C and Q_D as shown, we can view the binary coded decimal count as it takes place.



Shift Registers:- Flip flops can be used to store a single bit of binary data (1 or 0). A **Register** is a device which is used to store such information.

The information stored within these registers can be transferred with the help of **shift registers**. Shift Register is a group of flip flops used to store **multiple bits** of data. The bits stored in such registers can be made to move within the registers and in/out of the registers by applying clock pulses.

The registers which will shift the bits to left are called “Shift left registers”.

Shift registers are basically of 4 types. These are:

1. Serial In Serial Out shift register
2. Serial In parallel Out shift register
3. Parallel In Serial Out shift register
4. Parallel In parallel Out shift register

1. Serial-In Serial-Out Shift Register (SISO):- The shift register, which allows serial input and produces a serial output is known as **Serial-In Serial-Out shift register**.

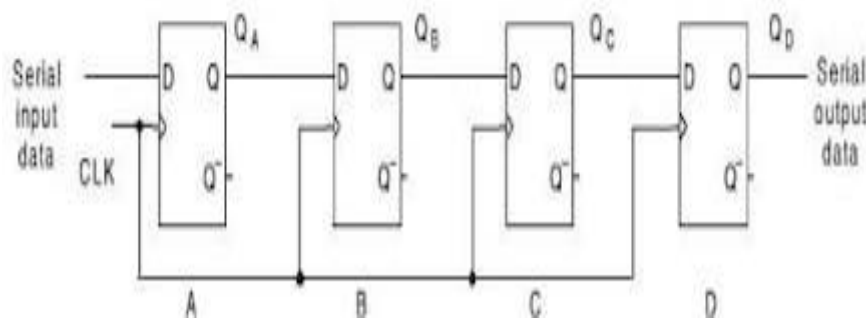
There are two types of shift registers.

- I. Serial right shift register.
- II. Serial left shift register.

I- Serial In Serial right shift Out Operation:- Data is shifted in the right-hand direction one bit at a time with each transition of the clock signal. The data enters

the shift register serially from the left hand side and after four clock transitions the 4-bit register has 4 bits of data. The data is shifted out serially one bit at a time from the right hand side of the register if clock signals are continuously applied. Thus after

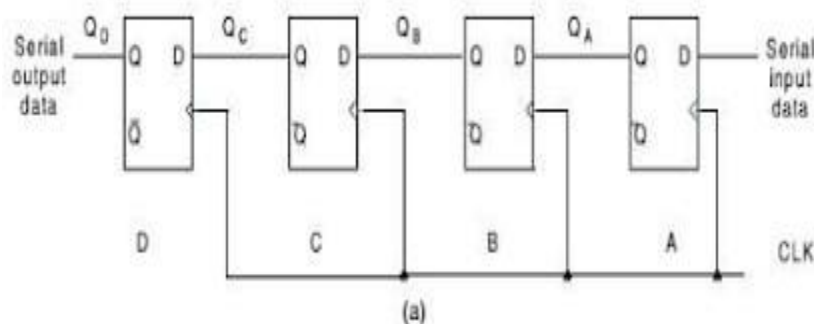
8 clock signals the 4-bit data is completely shifted out of the shift register.



II-Serial In Left shift Serial Out Operation:- Data is shifted in the left-

hand direction one bit at a time with each transition of the clock signal. The data enters the shift register serially from the right hand side and after four clock transitions the 4-bit register has 4-bits of data. The data is shifted out serially one bit at a time from the left hand side of the register if clock signals are continuously applied.

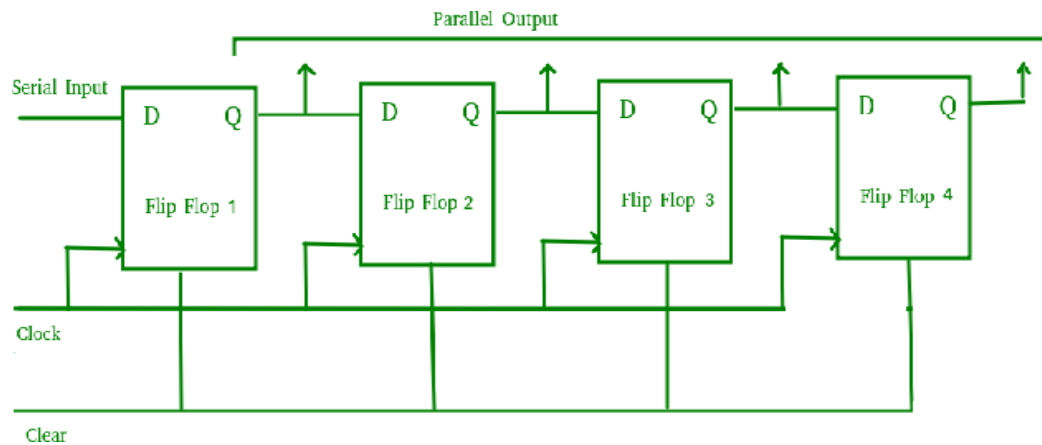
Thus after 8 clock signals the 4-bit data is completely shifted out of the shift register.



2.Serial-In Parallel-Out shift Register (SIPO) :-The shift register, which allows serial input and produces a parallel output is known as Serial-In Parallel-Out shift register.

The logic circuit given below shows a serial-in-parallel-out shift register. The circuit consists of four D flip-flops which are connected. The clear (CLR) signal is connected in addition to the clock signal to all the 4 flip flops in order to RESET them. The output of the first flip flop is connected to the input of the next flip flop and so on. All these flip-flops are synchronous with each other since the same clock signal is applied to each flip flop.

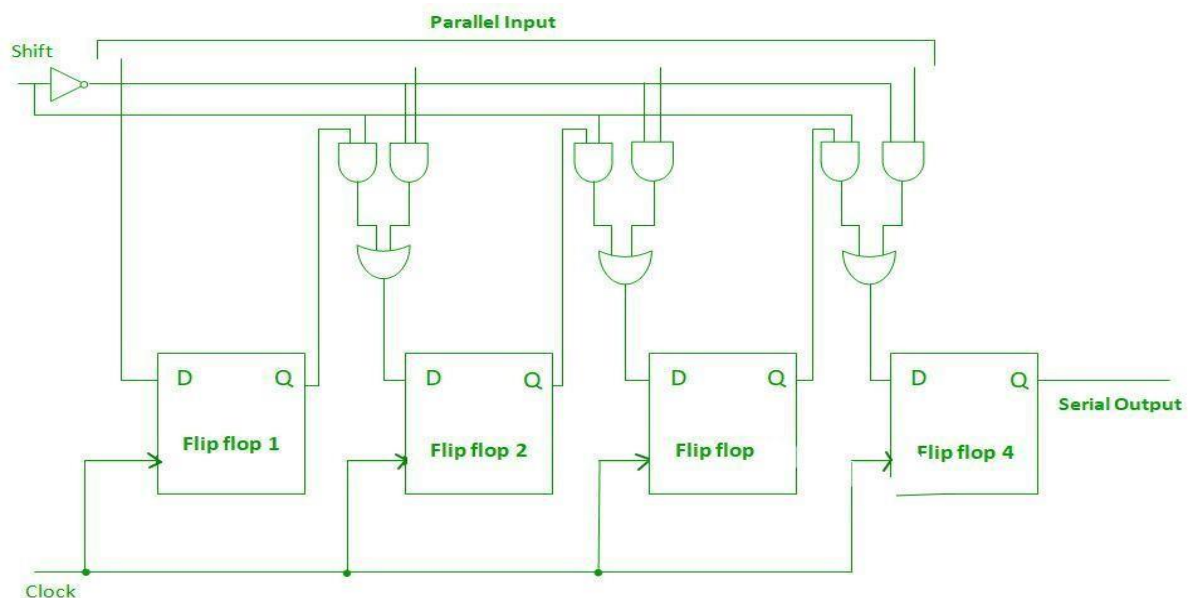
Since In logic circuit data given in serial and o/p observed in parallel.



3.Parallel-In Serial-Out Shift Register (PISO):- The shift register, which allows parallel data input and produces a serial output is known as **Parallel-In Serial-Out shift register**

The logic circuit given below shows a parallel-in-serial-out shift register. The circuit consists of four D flip-flops which are connected. The clock input is directly connected to all the flip flops but the input data is connected individually to each flip flop through a multiplexer at the input of every flip flop. The output of the previous flip flop and parallel data input are connected to the input of the MUX and the output of MUX is connected to the next flip flop. All these flip-flops are synchronous with each other since the same clock signal is applied to each flip flop.

Since In logic circuit data given in parallel and o/p observed in serial.

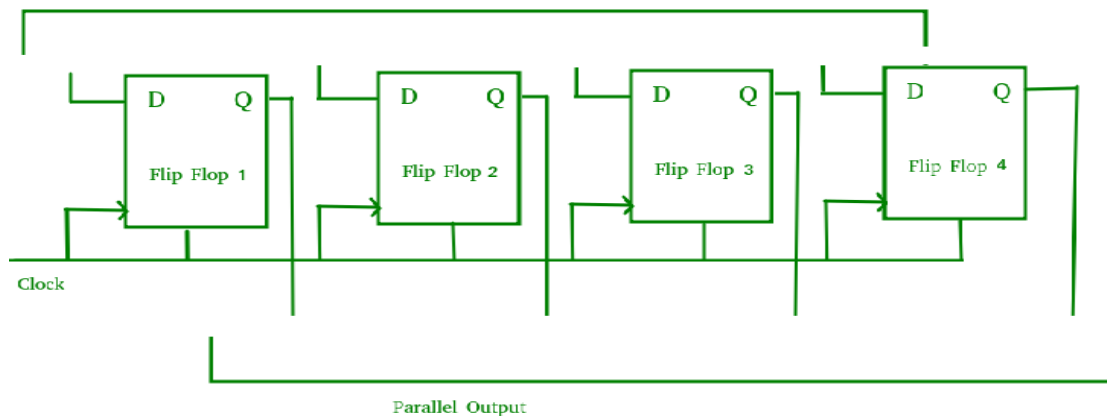


4. Parallel-In Parallel-Out Shift Register (PIPO):- The shift register, which allows parallel input and also produces a parallel output is known as Parallel-In parallel-Out shift register.

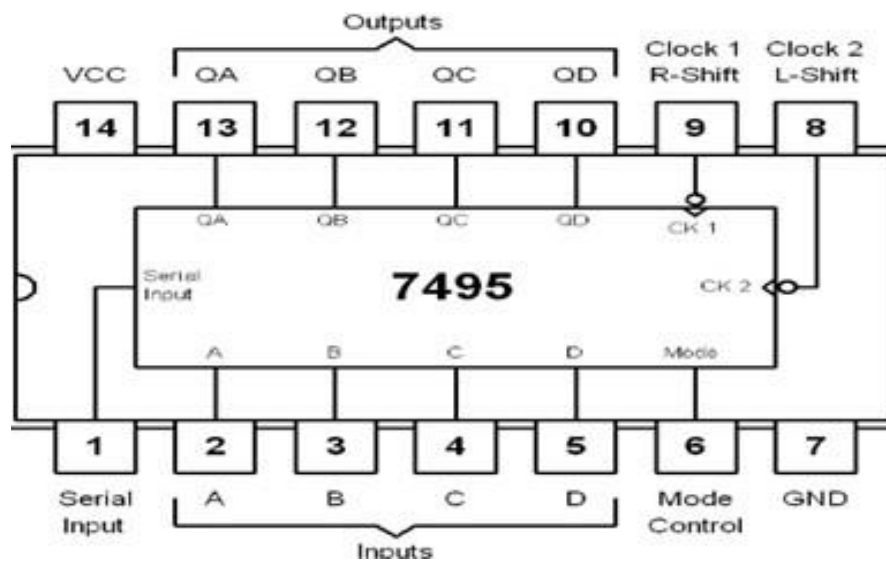
The logic circuit given below shows a parallel-in-parallel-out shift register. The circuit consists of four D flip-flops which are connected. The clear (CLR) signal and clock signals are connected to all the 4 flip flops. In this type of register, there are no interconnections between the individual flip-flops since no serial shifting of the data is required. Data is given as input separately for each flip flop and in the same way, output also collected individually from each flip flop.

Since In logic circuit data given in parallel and o/p observed in parallel.

Parallel I/P



IC 7495:- It is a TTL IC working on 5V power supply. It is a universal shift register of 4 bits. It can be operated in all the 4 modes. Provisions are there for entering data serially and parallel. With the help of mode control serial entry or parallel entry is possible. There are two clock inputs, one controlling right shift and the other, the left shift and parallel in parallel out mode of data transfer. The functional pin configuration of IC 7495 is as follows



However some external connections must be done to operate the shift register in any of the 4 modes.

Date: / /

Ref. Logic Gates.

OR - $Y = A + B$ AND $Y = A \cdot B$ NOT $Y = \overline{A}$	NOR $Y = \overline{A + B}$ NAND $Y = \overline{A \cdot B}$ XOR $Y = A \oplus B = \overline{A} \cdot B + A \cdot \overline{B}$ XNOR $Y = \overline{A \oplus B} = \overline{A \cdot B + A \cdot \overline{B}}$
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